UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*2639108572*

**BIOLOGY** **5090/02**

Paper 2 Theory **May/June 2009**

**1 hour 45 minutes**

Candidates answer on the Question Paper.

No Additional Materials are required.

**READ THESE INSTRUCTIONS FIRST**

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

**Section A**
Answer **all** questions.
Write your answers in the spaces provided on the Question Paper.

**Section B**
Answer **all** the questions including questions 6, 7 and 8 **Either** or 8 **Or**.
Write your answers in the spaces provided on the Question Paper.
Write an **E** (for Either) or an **O** (for Or) next to the number 8 in the Examiner's grid below to indicate which question you have answered.

You are advised to spend no longer than one hour on Section A and no longer than 45 minutes on Section B.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

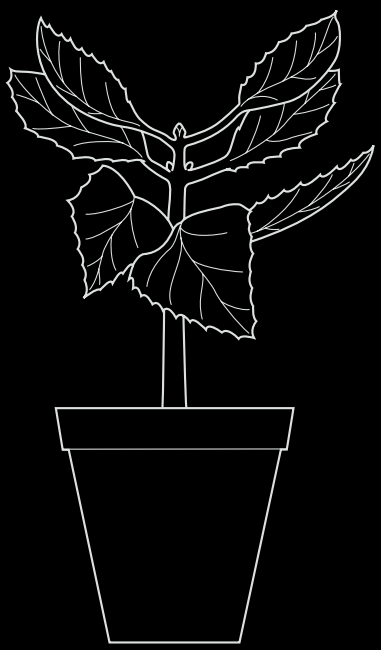
| For Examiner's Use | |
|---|---|
| Section A | |
| Section B | |
| 6 | |
| 7 | |
| 8 | |
| Total | |

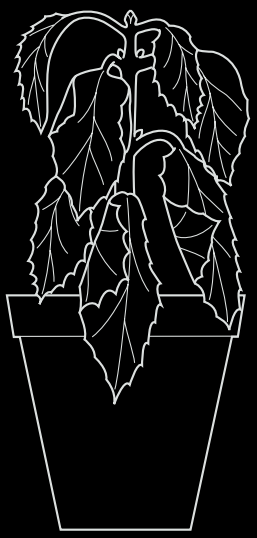
This document consists of **15** printed pages and **1** blank page.


© UCLES 2009

UNIVERSITY of CAMBRIDGE International Examinations

[Turn over





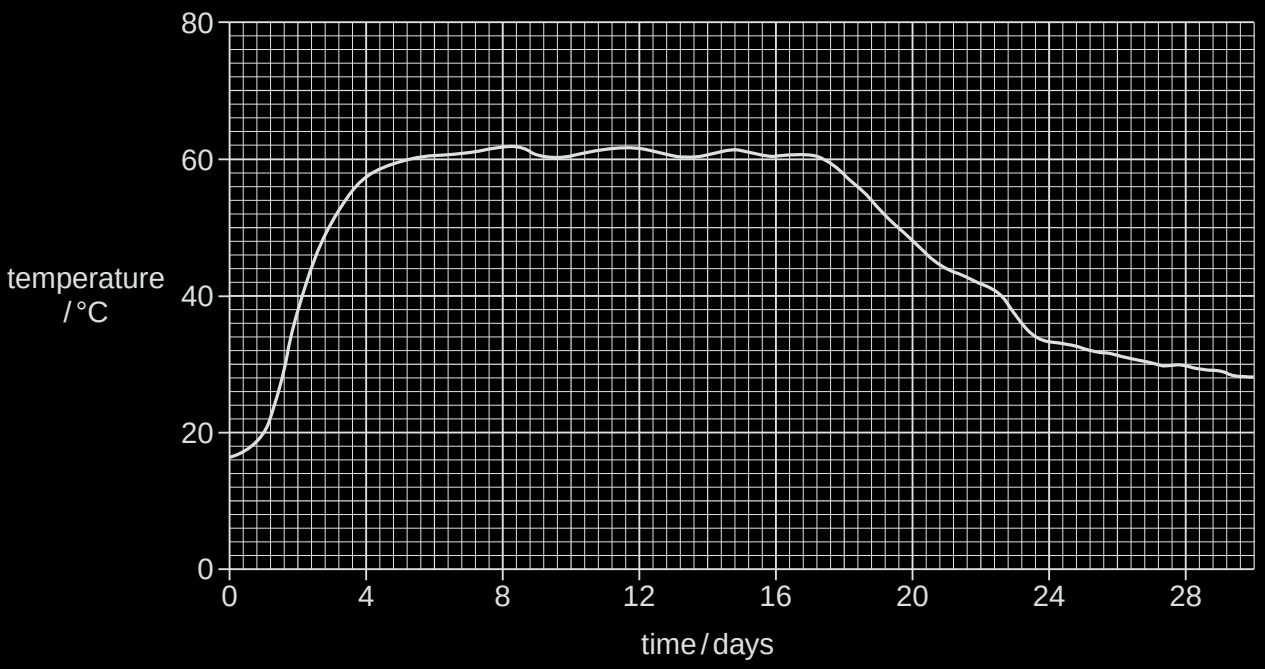



**Fig. 3.1**

**(a)** **(i)** Name two different types of microorganisms that could cause the changes of temperature shown in Fig. 3.1.

1. .....

2. ..... [2]

**(ii)** Name a type of microorganism that could **not** be responsible for the temperature changes and give a reason for your answer.

microorganism .....

reason ..... [2]

**(b)** Name a chemical that would be found in the compost in higher concentration at day 30 than at day 1 and explain your answer.

chemical .....

explanation ..... [2]

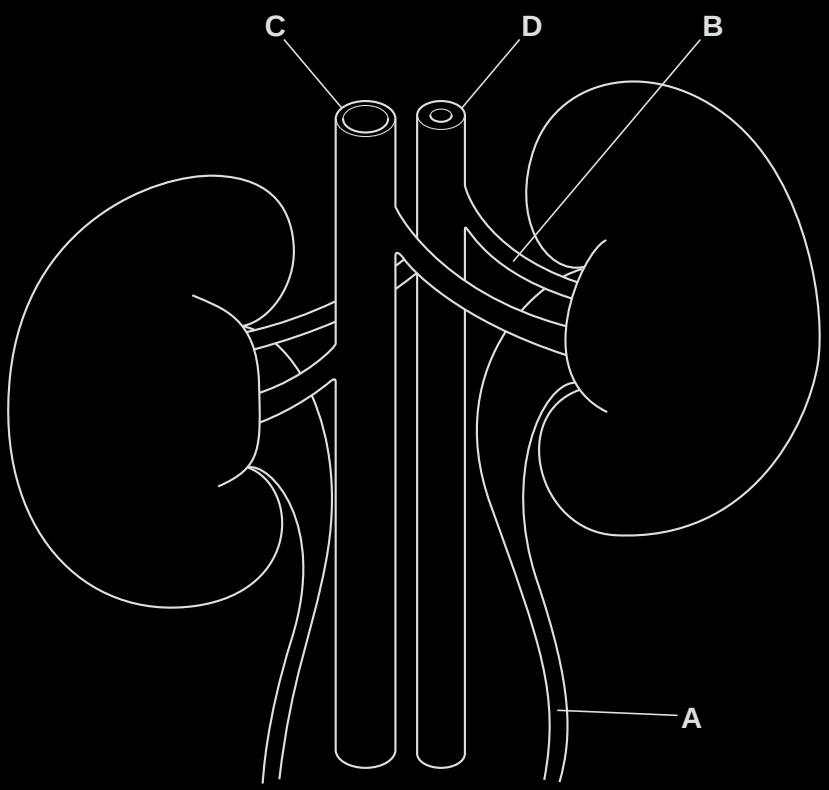



**Fig. 4.1**

**(a)** **(i)** Identify structure **A** in Fig. 4.1.

..... [1]

**(ii)** Peristalsis occurs continually in structure **A**. Describe and explain how this helps the structure to carry out its function.

.....

.....

..... [3]

**(b)** Identify structure **B** on Fig. 4.1 and state how the structural features of **C** and **D** enabled you to make your identification.

structure **B** .....

structural features of **C** and **D** .....

..... [3]

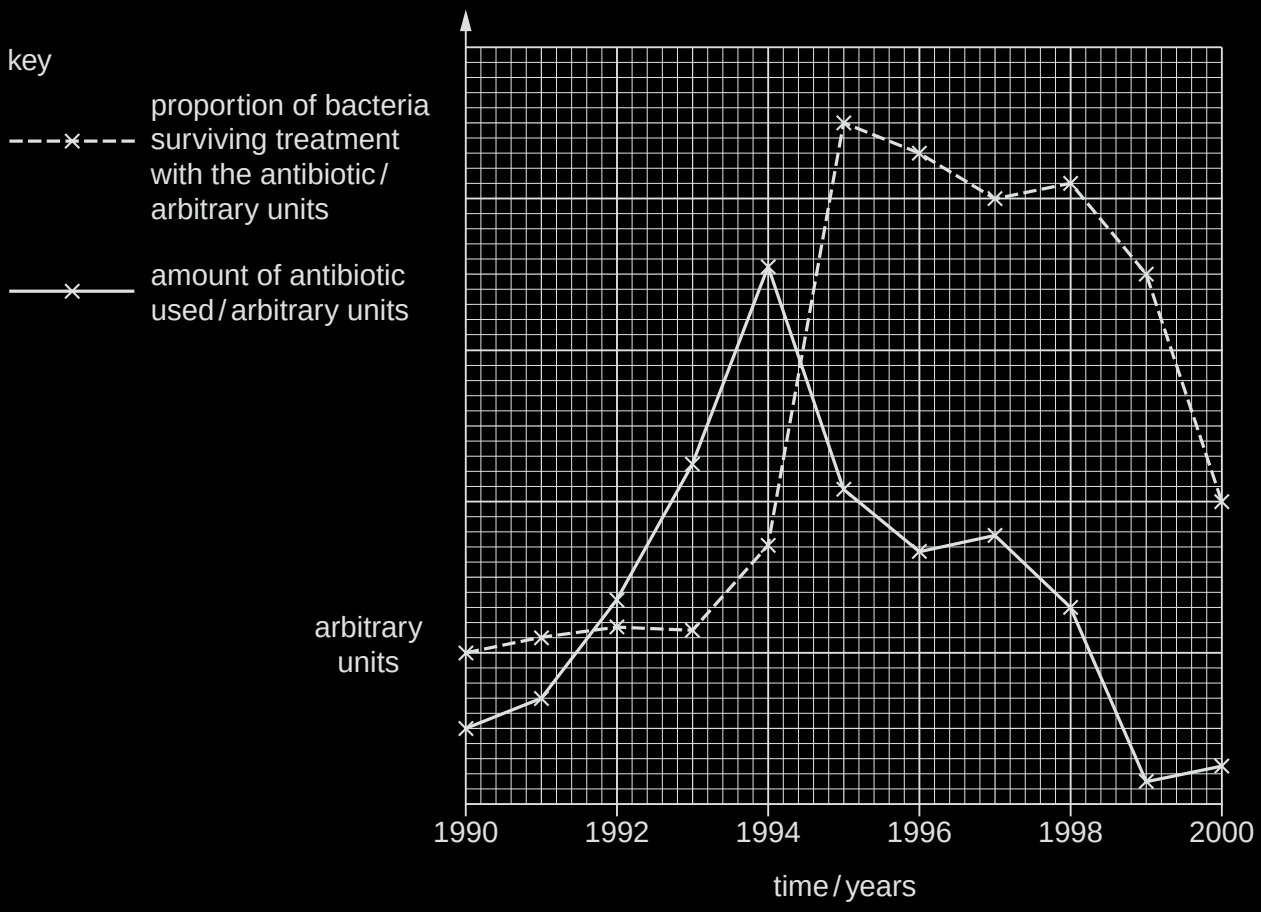



**Fig. 5.1**

**(a)** Name an antibiotic.

.....  [1]

**(b)** State the period of time during which the antibiotic was most effective at treating the infection in the hospital.

.....  [1]

**(c)** Suggest and explain possible causes for the increase in the proportion of bacteria that survived treatment with the antibiotic after 1994.

.....

.....

.....

.....

.....  [5]